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Voyager IQ — Design System

> **Version 1.0 · 3 November 2026** Linh Long Engineering / Rocket EMS · repository `ll-poc` **Source of truth:** `src/theme/` · **Living specimen:** `/design-system`

Everything in this document was read out of a working build. Every number was taken from the source; every picture was captured from the running application at 2× on 3 November 2026. Where a figure came from measuring the screen rather than from a rule, it says so.

Part A — Start here

A1 · What this is

The file someone needs to **build** a new Voyager IQ screen without asking, and the file a reviewer needs to **tell a correct screen from a plausible one**.

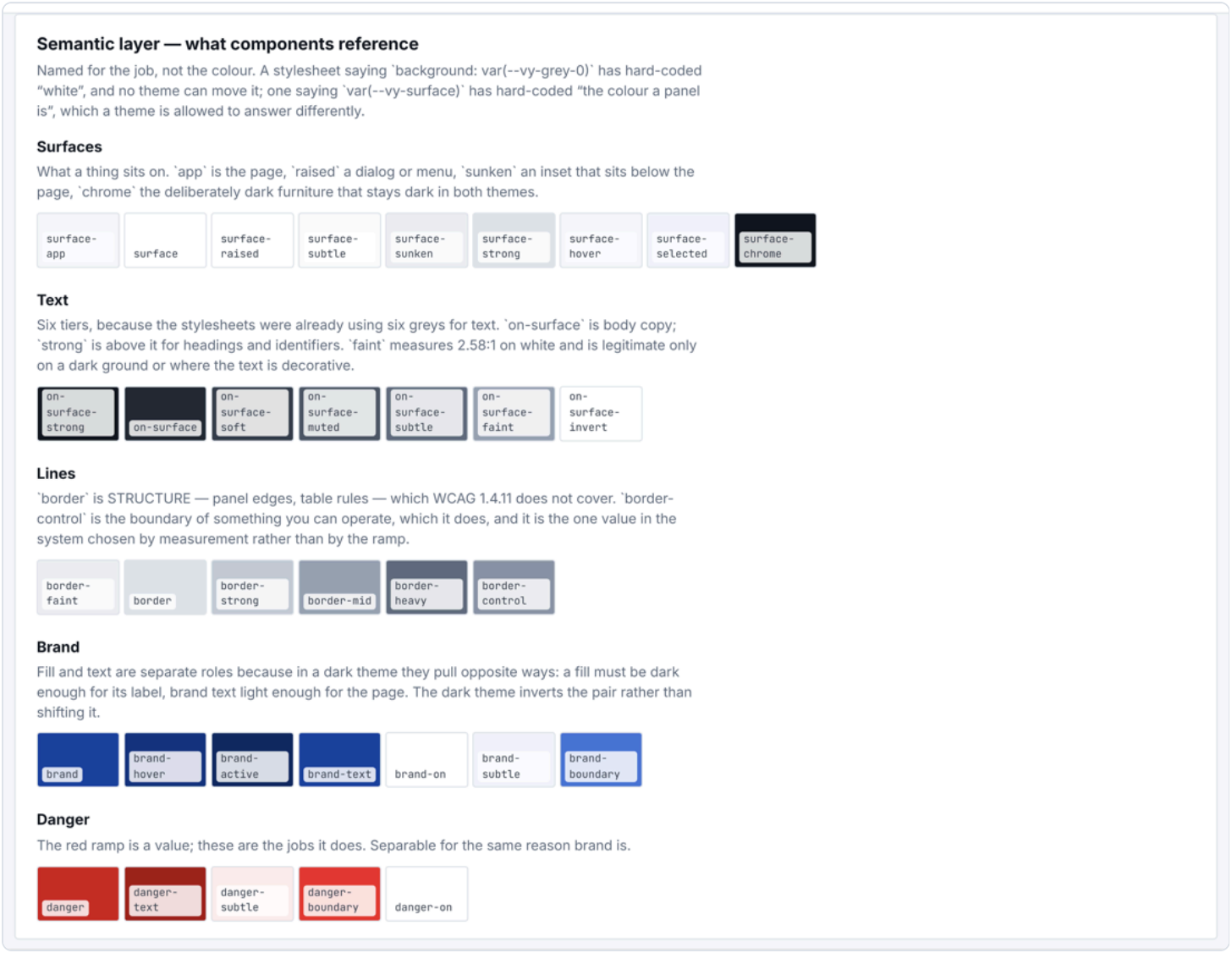
It is not a style guide for a drawing tool. It describes a system that is running: the tokens are CSS custom properties, the components are React, and the specimen page renders the real thing.

A2 · The three layers

Layer	What it is	Where	May a screen use it?
1 · Ramps	The raw palette. 10 blues, 12 greys, 4 reds	`tokens.css`	**No**
2 · Semantic roles	`surface`, `on-surface`, `brand`, `danger` … named for the job	`tokens.css`	**Yes — only this**
3 · Components	Button, Field, Grid, Dialog…	`src/ui`, `src/components`	Yes

A stylesheet saying `background: var(--vy-grey-0)` has hard-coded *white*, and no theme can move it. One saying `var(--vy-surface)` has hard-coded *"the colour a panel is"*, which a theme is allowed to answer differently.

Layer 1 exists so that layer 2 can be consistent. It is not for use in a component.



A3 · Five rules before you build

1. **Is there a token?** If yes, use it. If no, add one — and say in the code why the scale did not already have it.
2. **Do not invent content.** Field names, option lists, status words and behaviour are facts about the business, not design decisions. Look them up before drawing them.
3. **Both themes, both densities, 820px.** A screen that has only been seen in light mode on a 1920px monitor has been seen once.

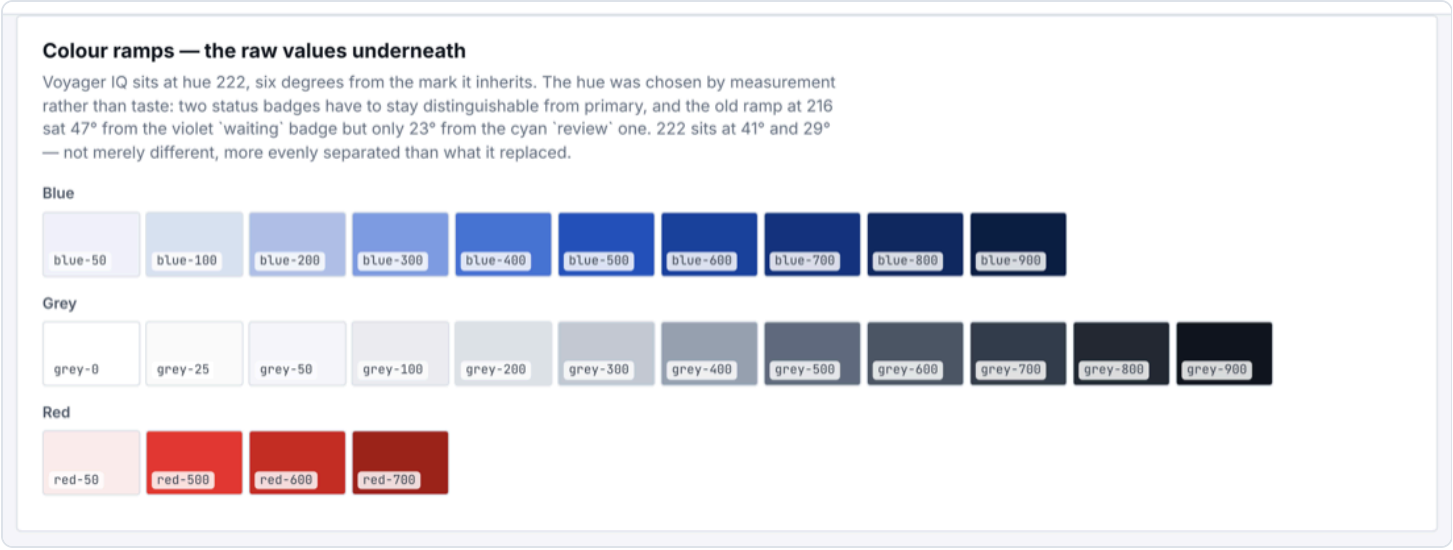
4. **A magic number carries a note.** A width that deviates from its role needs a `widthNote` saying what was measured.
5. **Say what is not real.** A derived figure, an unbuilt action and a fake send are each labelled where they appear, not only in a commit message.

Part B — Foundations

B1 · Colour

B1.1 The ramps *(layer 1 — not for use in components)*

Three ramps and nothing else. A fourth colour in a screen is a bug unless it is a status, and the statuses are in B1.6.



Token	Value
`--vy-blue-50`	`#f1f4fa`
`--vy-blue-100`	`#dae2f3`
`--vy-blue-200`	`#b0c2ea`
`--vy-blue-300`	`#7f9ce1`
`--vy-blue-400`	`#4973d4`
`--vy-blue-500`	`#2754bc`
`--vy-blue-600`	`#1a419e`
`--vy-blue-700`	`#15347e`
`--vy-blue-800`	`#112960`
`--vy-blue-900`	`#0d1e44`
`--vy-red-50`	`#fdeceb`
`--vy-red-500`	`#e23a32`
`--vy-red-600`	`#c62d26`
`--vy-red-700`	`#9e231d`
`--vy-grey-0`	`#ffffff`
`--vy-grey-25`	`#fbfcfd`
`--vy-grey-50`	`#f5f7fa`
`--vy-grey-100`	`#eceff3`
`--vy-grey-200`	`#dde2e9`
`--vy-grey-300`	`#c3cbd6`
`--vy-grey-400`	`#97a2b2`
`--vy-grey-500`	`#626d7e`
`--vy-grey-600`	`#4d5766`

Token	Value
`--vy-grey-700`	`#363e4b`
`--vy-grey-800`	`#232a34`
`--vy-grey-900`	`#131820`

B1.2 Surfaces — what a thing sits on

`app` is the page · `raised` a dialog or menu · `sunken` an inset below the page · `chrome` the deliberately dark furniture that stays dark in **both** themes.

Token	Value	Resolves to	Dark
`--vy-surface-app`	`var(--vy-grey-50)`	`#f5f7fa`	`#0f131a`
`--vy-surface`	`var(--vy-grey-0)`	`#ffffff`	`#161c25`
`--vy-surface-raised`	`var(--vy-grey-0)`	`#ffffff`	`#1e2531`
`--vy-surface-subtle`	`var(--vy-grey-25)`	`#fbfcfd`	`#131922`
`--vy-surface-sunken`	`var(--vy-grey-100)`	`#eceff3`	`#0b0e14`
`--vy-surface-strong`	`var(--vy-grey-200)`	`#dde2e9`	`#2a323f`
`--vy-surface-hover`	`var(--vy-grey-50)`	`#f5f7fa`	`#1e2531`
`--vy-surface-selected`	`var(--vy-blue-50)`	`#f1f4fa`	`#1b2a4a`
`--vy-surface-chrome`	`var(--vy-grey-900)`	`#131820`	`#1e2531`
`--vy-surface-chrome-soft`	`var(--vy-grey-700)`	`#363e4b`	`#2a323f`

B1.3 Text colour — six tiers

`on-surface` is body copy; `strong` sits above it for headings and identifiers.

> ⚠️ `on-surface-faint` **measures 2.58:1 on white.** Legitimate only on a dark ground or where the text is decorative — never for anything a user has to read. See E1.

Token	Value	Resolves to	Dark
`--vy-on-surface-strong`	`var(--vy-grey-900)`	`#131820`	`#f5f8fc`
`--vy-on-surface`	`var(--vy-grey-800)`	`#232a34`	`#e9edf3`
`--vy-on-surface-soft`	`var(--vy-grey-700)`	`#363e4b`	`#d3dae4`
`--vy-on-surface-muted`	`var(--vy-grey-600)`	`#4d5766`	`#aab4c3`
`--vy-on-surface-subtle`	`var(--vy-grey-500)`	`#626d7e`	`#8f9aaa`
`--vy-on-surface-faint`	`var(--vy-grey-400)`	`#97a2b2`	`#67727f`
`--vy-on-surface-invert`	`var(--vy-grey-0)`	`#ffffff`	`#0f131a`
`--vy-on-chrome-muted`	`var(--vy-grey-300)`	`#c3cbd6`	`#c3cbd6`
`--vy-on-dark-fg`	`#ffffff`	—	*(same)*
`--vy-on-dark-secondary`	`var(--vy-grey-300)`	`#c3cbd6`	*(same)*
`--vy-on-dark-muted`	`var(--vy-grey-400)`	`#97a2b2`	*(same)*
`--vy-on-dark-hairline`	`rgba(255, 255, 255, .07)`	—	*(same)*
`--vy-on-dark-hover`	`rgba(255, 255, 255, .05)`	—	*(same)*
`--vy-on-dark-active`	`rgba(43, 99, 184, .22)`	—	*(same)*

B1.4 Lines

`border` is **structure** — panel edges, table rules — which WCAG 1.4.11 does not cover. `border-control` is the boundary of something you can **operate**, which it does cover.

It is the one value in the system chosen by **measurement** rather than by the ramp. The sweep of 5 September found why: the grey ramp has **no step that conforms**. grey-400 `#97a2b2` is 2.58:1 against white and misses by 0.42; the next step overshoots to 5.24. `--vy-border-control` `#8a94a6` was introduced between them at **3.06:1** — the lightest value that conforms. It governs boundaries only: input, textarea, combobox, checkbox, radio.

Token	Value	Resolves to	Dark
`--vy-border-control`	`#8a94a6`	—	`#6d7889`
`--vy-border-faint`	`var(--vy-grey-100)`	`#eceff3`	`#212936`
`--vy-border`	`var(--vy-grey-200)`	`#dde2e9`	`#2a323f`
`--vy-border-strong`	`var(--vy-grey-300)`	`#c3cbd6`	`#3b4553`
`--vy-border-mid`	`var(--vy-grey-400)`	`#97a2b2`	`#55606f`
`--vy-border-heavy`	`var(--vy-grey-500)`	`#626d7e`	`#55606f`
`--vy-border-emphasis`	`2px`	—	*(same)*

B1.5 Brand and danger

Fill and text are **separate roles**, because in a dark theme they pull opposite ways: a fill must be dark enough for its label, brand text light enough for the page. The dark theme inverts the pair rather than shifting it.

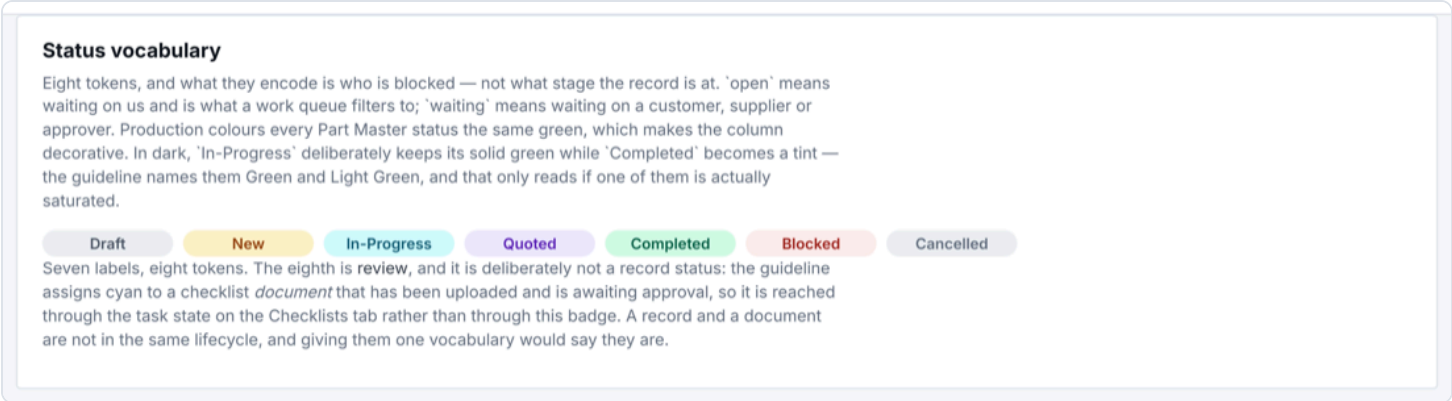
Token	Value	Resolves to	Dark
`--vy-brand`	`var(--vy-blue-600)`	`#1a419e`	`#7f9ce1`
`--vy-brand-hover`	`var(--vy-blue-700)`	`#15347e`	`#b0c2ea`
`--vy-brand-active`	`var(--vy-blue-800)`	`#112960`	`#dae2f3`
`--vy-brand-text`	`var(--vy-blue-600)`	`#1a419e`	`#7f9ce1`
`--vy-brand-text-hover`	`var(--vy-blue-700)`	`#15347e`	`#b0c2ea`
`--vy-brand-text-deep`	`var(--vy-blue-800)`	`#112960`	`#dae2f3`
`--vy-brand-on`	`var(--vy-grey-0)`	`#ffffff`	`#0f131a`
`--vy-brand-subtle`	`var(--vy-blue-50)`	`#f1f4fa`	`#1b2a4a`
`--vy-brand-subtle-strong`	`var(--vy-blue-100)`	`#dae2f3`	`#223357`
`--vy-brand-light`	`var(--vy-blue-400)`	`#4973d4`	`#7f9ce1`
`--vy-brand-soft`	`var(--vy-blue-500)`	`#2754bc`	`#4973d4`
`--vy-brand-boundary`	`var(--vy-blue-400)`	`#4973d4`	`#4973d4`
`--vy-brand-boundary-soft`	`var(--vy-blue-200)`	`#b0c2ea`	`#2754bc`
`--vy-brand-boundary-light`	`var(--vy-blue-300)`	`#7f9ce1`	`#7f9ce1`
`--vy-brand-boundary-strong`	`var(--vy-blue-500)`	`#2754bc`	`#4973d4`

Token	Value	Resolves to	Dark
`--vy-danger`	`var(--vy-red-600)`	`#c62d26`	`#c62d26`
`--vy-danger-on`	`var(--vy-grey-0)`	`#ffffff`	`#ffffff`
`--vy-danger-text`	`var(--vy-red-700)`	`#9e231d`	`#f0918a`
`--vy-danger-text-deep`	`var(--vy-red-700)`	`#9e231d`	`#f5aba5`
`--vy-danger-subtle`	`var(--vy-red-50)`	`#fdeceb`	`#38201e`
`--vy-danger-boundary`	`var(--vy-red-500)`	`#e23a32`	`#e0655c`

B1.6 Status vocabulary

Eight statuses, each a foreground/background pair.

> **A status is a word and a colour, never a colour alone.** The pair is what makes it legible to someone who cannot separate the hues.



Token	Value	Resolves to	Dark
`--vy-status-draft-fg`	`var(--vy-grey-600)`	`#4d5766`	`#aab4c3`
`--vy-status-draft-bg`	`var(--vy-grey-100)`	`#eceff3`	`#2a323f`
`--vy-status-open-fg`	`#92400e`	—	`#eeb796`
`--vy-status-open-bg`	`#fef3c7`	—	`#38281e`
`--vy-status-progress-fg`	`#155e75`	—	*(same)*
`--vy-status-progress-bg`	`#cffafe`	—	*(same)*
`--vy-status-done-fg`	`#065f46`	—	`#96eed5`
`--vy-status-done-bg`	`#d1fae5`	—	`#1e3831`
`--vy-status-waiting-fg`	`#5b21b6`	—	`#b896ee`
`--vy-status-waiting-bg`	`#ede9fe`	—	`#281e38`
`--vy-status-review-fg`	`#0e7490`	—	`#96dbee`
`--vy-status-review-bg`	`#ecfeff`	—	`#1e3338`
`--vy-status-blocked-fg`	`var(--vy-red-700)`	`#9e231d`	`#ee9a96`
`--vy-status-blocked-bg`	`var(--vy-red-50)`	`#fdeceb`	`#38201e`
`--vy-status-cancelled-fg`	`var(--vy-grey-500)`	`#626d7e`	`#aab4c3`
`--vy-status-cancelled-bg`	`var(--vy-grey-100)`	`#eceff3`	`#2a323f`
`--vy-status-slot-w`	`104px`	—	*(same)*

B1.7 Row tones and accents

A tone is a row's **state in the quote**, not a severity. Three of them paint most of a BoM grid, and the legend under the grid names all three.

Token	Value	Resolves to	Dark
`--vy-tone-excluded-bg`	`var(--vy-grey-100)`	`#eceff3`	`#2a323f`
`--vy-tone-excluded-fg`	`var(--vy-grey-500)`	`#626d7e`	`#aab4c3`
`--vy-tone-missing-bg`	`var(--vy-red-50)`	`#fdeceb`	`#38201e`
`--vy-tone-missing-fg`	`var(--vy-red-700)`	`#9e231d`	`#ee9a96`
`--vy-tone-short-bg`	`var(--vy-status-open-bg)`	`#fef3c7`	*(same)*
`--vy-tone-covered-bg`	`var(--vy-status-done-bg)`	`#d1fae5`	*(same)*
`--vy-tone-known-mfg-bg`	`var(--vy-status-open-bg)`	`#fef3c7`	*(same)*
`--vy-tone-pending-bg`	`var(--vy-surface-sunken)`	`#eceff3`	*(same)*

Token	Value	Resolves to	Dark
`--vy-accent-positive`	`#047857`	—	`#75f0cd`
`--vy-accent-caution`	`#d97706`	—	`#f0b775`
`--vy-accent-warning`	`#b45309`	—	`#f0aa75`
`--vy-accent-waiting`	`#6d28d9`	—	`#a679ec`
`--vy-accent-neutral`	`#4d5766`	—	`#aab4c3`
`--vy-accent-bar-w`	`3px`	—	*(same)*

B1.8 How dark mode is wired

Three blocks, and **a token must appear in all three** or the two ways of choosing dark disagree:

Block	Who it ser
`:root { … }`	Light — the default
`:root[data-theme="dark"] { … }`	A user who **picked** dar in the user menu
`@media (prefers-color-scheme: dark) { :root:not([data-theme="light"]) { … } }`	A user who **OS** is dark

> This has already failed once: four tokens existed only in the media copy, so a user who picked dark in the menu got grey-400 borders and a near-white priority dot on a near-black ground. **Adding a colour token means editing three places.**

The same two blocks in the dark theme. Every role keeps its job and changes its value — and the status pairs stay legible rather than being inverted mechanically:

Semantic layer — what components reference

Named for the job, not the colour. A stylesheet saying `background: var(--vy-grey-0)` has hard-coded “white”, and no theme can move it; one saying `var(--vy-surface)` has hard-coded “the colour a panel is”, which a theme is allowed to answer differently.

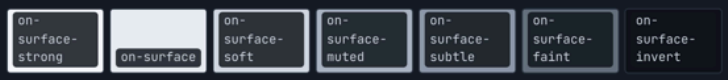
Surfaces

What a thing sits on. `app` is the page, `raised` a dialog or menu, `sunken` an inset that sits below the page, `chrome` the deliberately dark furniture that stays dark in both themes.



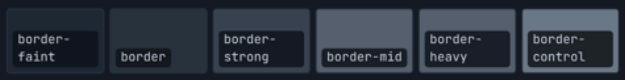
Text

Six tiers, because the stylesheets were already using six greys for text. `on-surface` is body copy; `strong` is above it for headings and identifiers. `faint` measures 2.58:1 on white and is legitimate only on a dark ground or where the text is decorative.



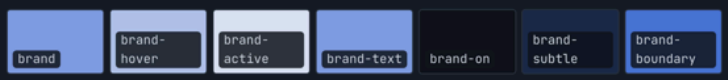
Lines

`border` is STRUCTURE — panel edges, table rules — which WCAG 1.4.11 does not cover. `border-control` is the boundary of something you can operate, which it does, and it is the one value in the system chosen by measurement rather than by the ramp.



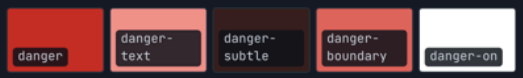
Brand

Fill and text are separate roles because in a dark theme they pull opposite ways: a fill must be dark enough for its label, brand text light enough for the page. The dark theme inverts the pair rather than shifting it.



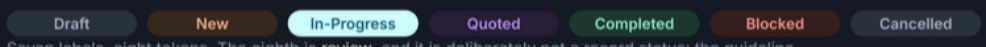
Danger

The red ramp is a value; these are the jobs it does. Separable for the same reason brand is.



Status vocabulary

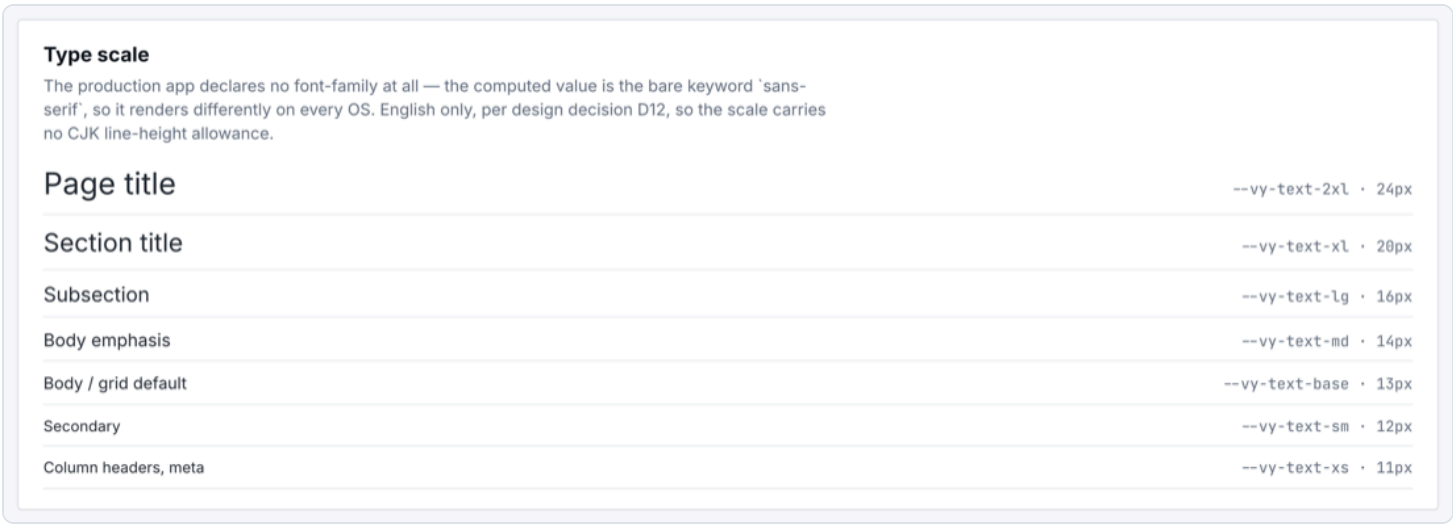
Eight tokens, and what they encode is who is blocked — not what stage the record is at. `open` means waiting on us and is what a work queue filters to; `waiting` means waiting on a customer, supplier or approver. Production colours every Part Master status the same green, which makes the column decorative. In dark, `In-Progress` deliberately keeps its solid green while `Completed` becomes a tint — the guideline names them Green and Light Green, and that only reads if one of them is actually saturated.



Seven labels, eight tokens. The eighth is **review**, and it is deliberately not a record status: the guideline assigns cyan to a checklist *document* that has been uploaded and is awaiting approval, so it is reached through the task state on the Checklists tab rather than through this badge. A record and a document are not in the same lifecycle, and giving them one vocabulary would say they are.

B2 · Typography

Two families. **Inter Variable** for everything; **JetBrains Mono** for identifiers that get read aloud or compared character by character — part numbers, MPNs, error ids. Body text is never mono — at 13px it is harder to read, and nothing about a sentence needs a fixed advance width.



Token	Value
`--vy-font-sans`	`"Inter Variable", "Inter", system-ui, -apple-system, "Segoe UI",`
`--vy-font-mono`	`"JetBrains Mono", ui-monospace, "SF Mono", "Cascadia Mono", Consc`
`--vy-text-2xs`	`12px`
`--vy-text-xs`	`13px`
`--vy-text-sm`	`14px`
`--vy-text-base`	`15px`
`--vy-text-md`	`16px`
`--vy-text-lg`	`18px`
`--vy-text-xl`	`20px`
`--vy-text-2xl`	`24px`
`--vy-text-3xl`	`28px`
`--vy-weight-regular`	`400`
`--vy-weight-medium`	`500`
`--vy-weight-semibold`	`600`
`--vy-weight-bold`	`650`
`--vy-weight-heavy`	`700`

The scale shifts on a touch device. Under `@media (pointer: coarse)` the body range moves up by a constant **+2px** — 12, 13, 14, 15, 16, 18 instead of 10, 11, 12, 13, 14, 16 — which keeps every relationship intact and lifts the smallest step to a readable 12. Display sizes (`xl` and up) do not move: they are already comfortable, and growing a heading on a 375px screen buys nothing and costs a wrap.

Casing — three levels, drawn differently on purpose

Buttons, links, tabs, form and dialog titles, section and card titles, **field labels**, column headers, filter labels, menu items and status pills take **Title Case**, with prepositions (for, to, of, by, as, in, on, at, from, with, via, per, into) lowercase unless they lead. Prose is sentence case.

Level	Case	Size / weight	Example
Card header	Title Case	14px / 600	`Select Action` · `Continue from Drafts`
Divider inside a card	**UPPERCASE**, .05em tracking	12px / 600	`QUOTE STRATEGY` · `BOM OPTIONS`
Field label	Title Case	12px / 600	`Customer` · `Qty to Quote` · `Markup (%)`

B3 · Spacing

B3.1 The scale

Token	Value
`--vy-space-1`	`2px`
`--vy-space-2`	`4px`
`--vy-space-3`	`6px`
`--vy-space-4`	`8px`
`--vy-space-5`	`10px`
`--vy-space-6`	`12px`
`--vy-space-7`	`14px`
`--vy-space-8`	`16px`
`--vy-space-9`	`20px`
`--vy-space-10`	`24px`
`--vy-space-11`	`28px`
`--vy-space-12`	`32px`
`--vy-space-14`	`40px`

B3.2 Which gap goes between two controls

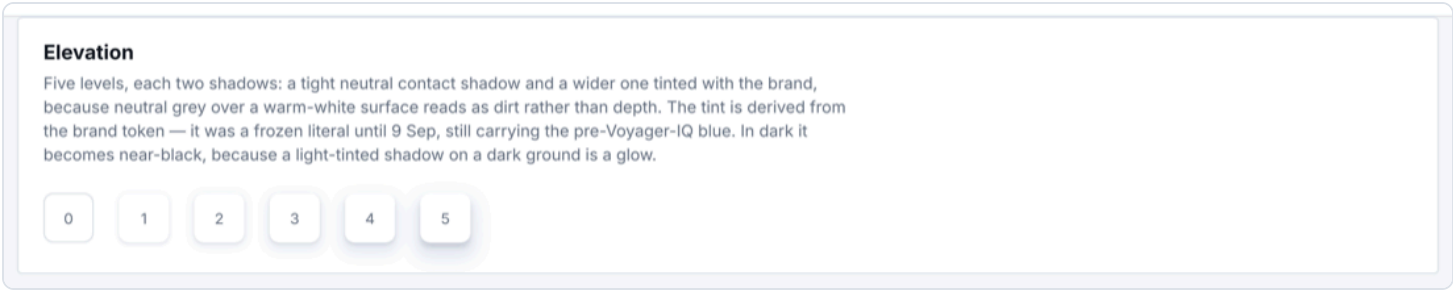
The scale does not say which step to pick, so the system states it. The tiers are what carry **grouping**, which is why they are not collapsed into one value:

Gap	Between	Examples
2px	Buttons in a cluster that reads as **one** control	dialog titlebar, pager arrows
4px	Icon buttons attached to the control they act on	a Select with Preview and Download
8px	**Two adjacent actions** — the default	`.vy-page-actions`, `.vy-run-nav-actions`, `.vy-cost-actions`
12px	Items that each hold a control **and** a label	filter checkboxes, which are 8px from their own words
16px	**Groups** on one row	toolbar filters / search / Columns

Two buttons side by side are 8px apart everywhere. A bigger gap means a real boundary, not a different author.

B4 · Shape, elevation, depth

Four radii, three shadows. Elevation is for things that **float** — a dialog, a menu, a toast. A card on a page gets a border, not a shadow.



Token	Value	Dark
`--vy-radius-sm`	`3px`	*(same)*
`--vy-radius-md`	`4px`	*(same)*
`--vy-radius-lg`	`6px`	*(same)*
`--vy-radius-pill`	`999px`	*(same)*
`--vy-shadow-sm`	`0 1px 2px rgba(19, 24, 32, .06)`	*(same)*
`--vy-shadow-md`	`0 2px 8px rgba(19, 24, 32, .10)`	*(same)*
`--vy-shadow-lg`	`0 8px 28px rgba(19, 24, 32, .14)`	*(same)*
`--vy-overlay-scrim`	`rgba(19, 24, 32, .45)`	`rgba(0, 0, 0, .62)`

Z-index — one flat list, so two overlays can never argue

Token	Value
`--vy-shell-sidebar-w`	`248px`
`--vy-shell-sidebar-w-collapsed`	`56px`
`--vy-shell-topbar-h`	`52px`
`--vy-z-drawer`	`10020`
`--vy-z-dialog`	`10040`
`--vy-z-menu`	`10070`
`--vy-z-palette`	`10075`
`--vy-z-toast`	`12500`
`--vy-z-skip`	`12600`
`--vy-dialog-md`	`640px`
`--vy-dialog-lg`	`960px`
`--vy-dialog-xl`	`1180px`

B5 · Sizing

B5.1 Control heights

Every control sits on **Kendo's own size scale**, configured rather than overridden — `_kendo-size.scss` sets the variables and the components inherit them. Forcing a height on a Kendo control instead is how a 29px select ends up beside a 34px input.

Token	Value
`--vy-row-h-compact`	`28px`
`--vy-row-h-comfortable`	`40px`
`--vy-row-h-relaxed`	`48px`
`--vy-target-min`	`44px`
`--vy-control-sm`	`28px`
`--vy-control-row`	`26px`
`--vy-control-md`	`34px`
`--vy-control-dense`	`32px`
`--vy-icon-sm`	`16px`

Token	Where it is used
`--vy-control-md` **34px**	Form inputs, selects, buttons — the default
`--vy-control-dense` **32px**	Toolbar controls: search, `size="sm"` buttons
`--vy-control-sm` **28px**	Compact density
`--vy-control-row` **26px**	An icon button **inside a grid row** — Kendo's `xs` step
`--vy-target-min` **44px**	A floor, under `pointer: coarse` **only**

> **44px is a touch rule, not a desktop one.** WCAG 2.5.5 is about a pointer target; applying it to a 27-field form on a mouse-driven screen adds ~270px of scrolling for nothing.

B5.2 Row density — a user preference

Token	Height	Setting
`--vy-row-h-compact`	28px	Compact
`--vy-row-h-comfortable`	40px	Comfortable *(default)*
`--vy-row-h-relaxed`	48px	Relaxed

B5.3 Column roles — a width is chosen once, by role

A grid column declares a **role**, not a pixel width.

Column roles

The rule that replaces uniform 108px columns. Width is a consequence of what the column holds — and a column empty in more than half its rows is hidden by default, with the reason stated in the chooser.

ROLE	WIDTH	RULE
ident	240px	The primary identifier. Monospaced, tabular, never truncated — it is how people refer to the record out loud and in email.
text	280px	The only role allowed to truncate. Fixed at 280px with the full value in a tooltip; the grid scrolls rather than squeezing every column.
code	96px	Short enumerations. Width is set to the longest member of the enum, so it never clips and never over-reserves.
number	104px	Right-aligned with tabular figures so magnitudes line up and a misplaced digit is visible.
money	124px	Right-aligned, currency-aware, always two decimals.
date	150px	Sized to the full rendered format. A truncated date is worse than no date — production truncates 100% of them.
datetime	190px	—
status	128px	One badge from the shared status tokens. Never free text.
priority	104px	—

Role	Width	What it holds
`ident`	240px	Part numbers, MPNs — must not truncate
`text`	280px	Descriptions and other prose
`code`	96px	Short codes, revisions
`number`	104px	Quantities
`money`	124px	Prices and amounts
`date`	150px	A day
`datetime`	190px	A moment — `09/09/2026 16:35:29`
`status`	128px	A status pill
`priority`	104px	A dot and a word

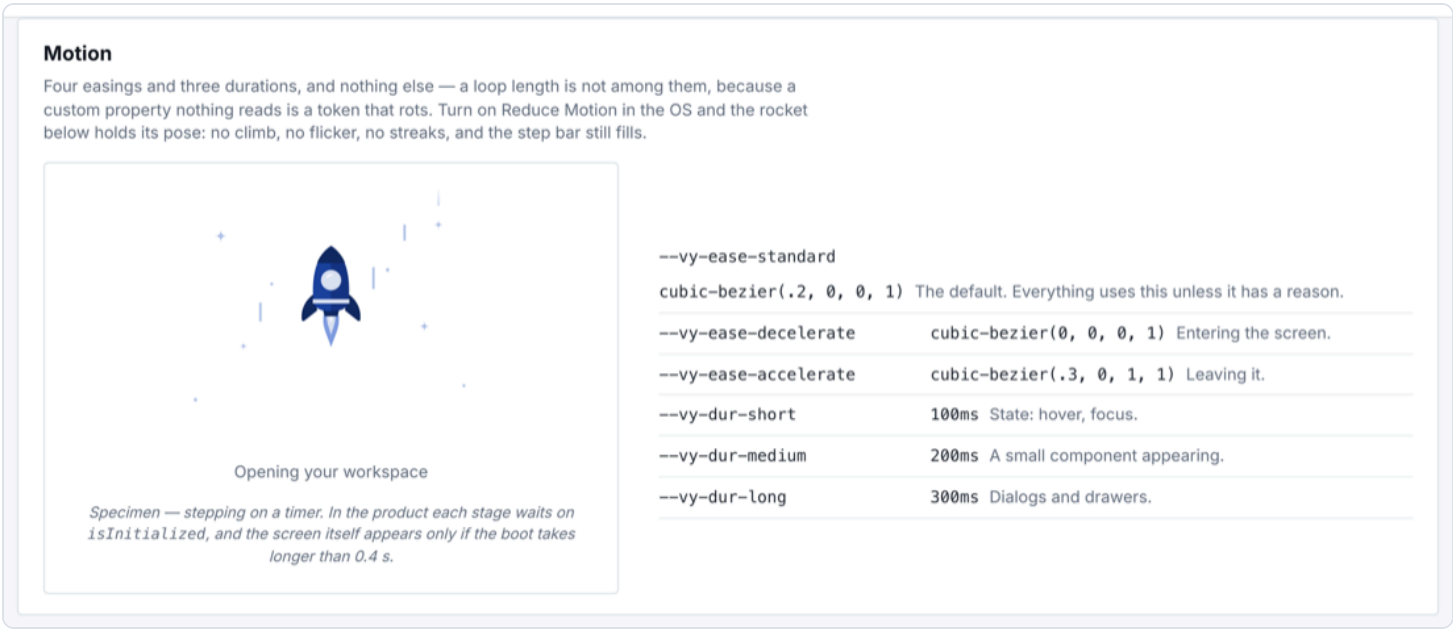
B5.4 Dialogs and shell

Size	Width	Opens maximised?	For
`md`	640px	No	A question: a confirmation, a short form
`lg`	960px	Yes *(overridable)*	Two blocks and a short table
`xl`	1180px	Yes	A work surface: a 21-column grid, a 4-step wizard

Shell token	Value
`--vy-shell-sidebar-w`	248px
`--vy-shell-sidebar-w-collapsed`	56px
`--vy-shell-topbar-h`	52px

B6 · Motion

Four easings and three durations, and nothing else. There is deliberately **no loop-length token**: a custom property nothing reads is a token that rots.



Token	Value	Use
`--vy-ease-standard`	`cubic-bezier(.2, 0, 0, 1)`	The default. Everything, unless it has a reason
`--vy-ease-emphasised`	`cubic-bezier(.2, 0, 0, 1)`	A moment that should be noticed. **Currently identical to** `standard` — named separately so the two can diverge without touching call sites
`--vy-ease-decelerate`	`cubic-bezier(0, 0, 0, 1)`	Something arriving
`--vy-ease-accelerate`	`cubic-bezier(.3, 0, 1, 1)`	Something leaving
`--vy-dur-short`	**100ms**	State changes: hover, focus
`--vy-dur-medium`	**200ms**	Small components entering
`--vy-dur-long`	**300ms**	Dialogs, drawers

The loading clock

Rule	Value
Show the boot screen only if the boot takes longer than	**400ms**
Exit — the craft leaves the top of the frame	**400ms**
"Still climbing…" appears after	**5s**
Minimum display time once shown	**none**

Holding a loading screen so it does not flash spends the user's time to make our indicator look better. Not showing one at all for a boot that beats the clock costs them nothing. The two rules are the two halves of one answer.

Reduced motion is a state, not a switch

Under `prefers-reduced-motion: reduce` the rocket **holds its pose** — no climb, no flicker, no streaks — and the step bar still fills. A welding arm with its animation removed is an arm with no sparks and an empty checklist, which says something different from what the state means, so the things that *fade in* are left on.

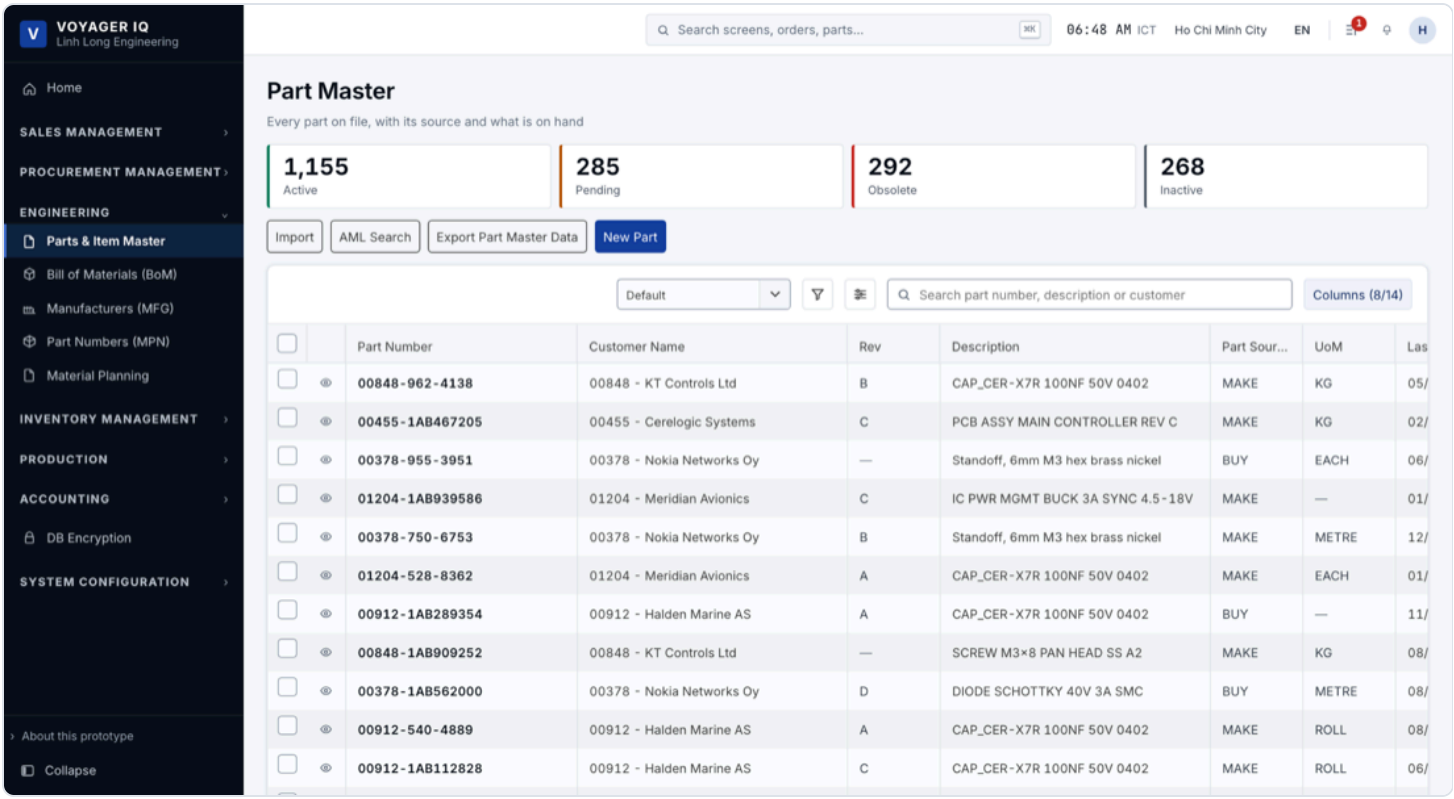
B7 · Responsive

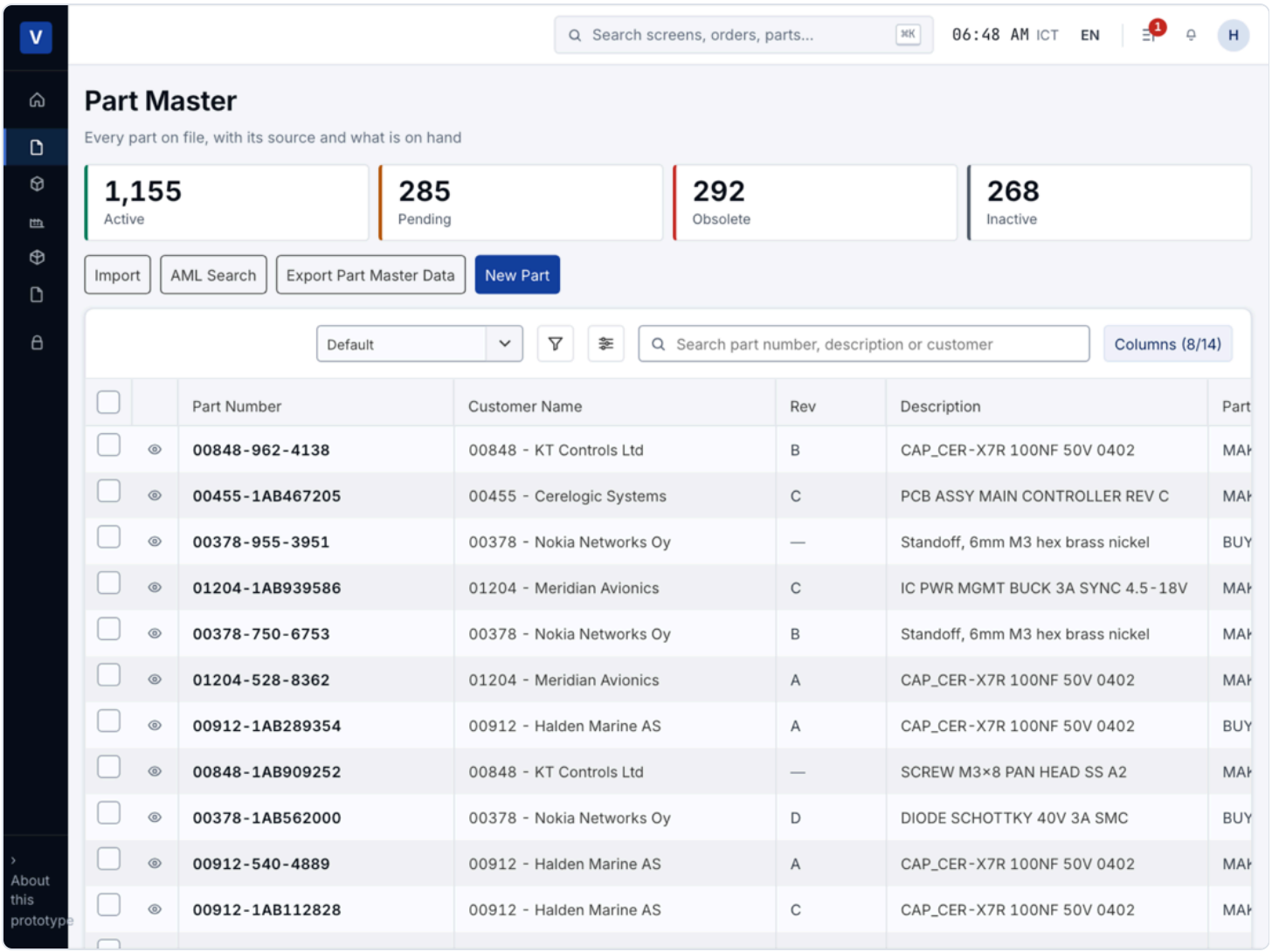
Breakpoints are raw pixels because `@media` cannot read custom properties — a documented exception to the no-magic-numbers rule.

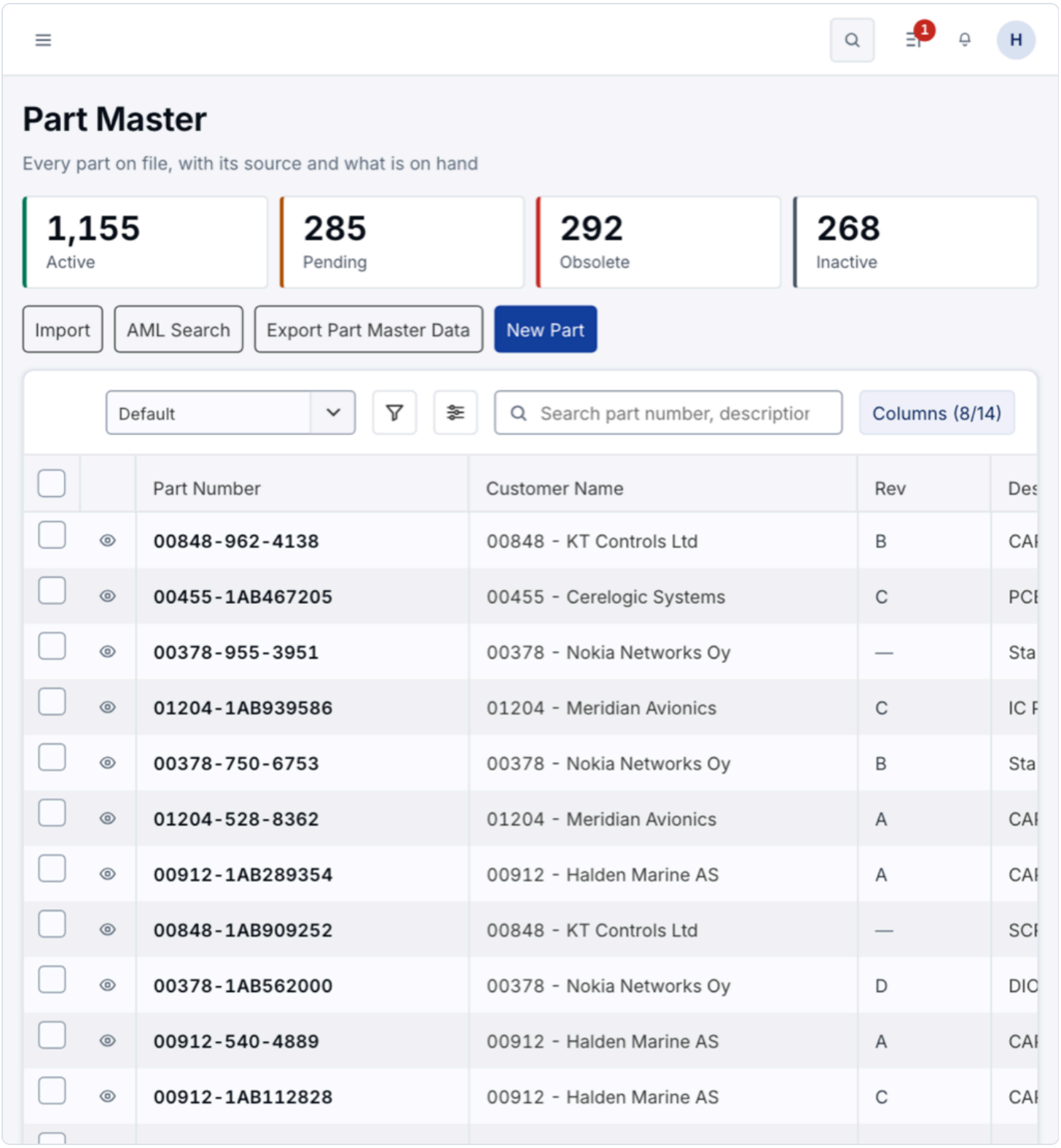
Width	What changes
≤ **1180px**	The 248px sidebar becomes a 56px icon rail
≤ **980px**	Two-column regions stack
≤ **820px**	The sidebar leaves the flow; a menu button reveals it
`pointer: coarse`	Type scale +2px on the body range; 44px target floor

The system runs on shop-floor terminals that are often 1366×768 or narrower. Below about 1100px a 248px sidebar stops being orientation and starts being the thing in the way — measured at 543px it was taking **46% of the viewport**.

The same screen at the three widths that matter. Full rail, icon rail, and no rail at all with a menu button in the top bar — and in all three the grid keeps its columns rather than shrinking them:




Search screens, orders, parts...
06:48 AM ICT
EN
H
Part Master
Every part on file, with its source and what is on hand
1,155
Active
285
Pending
292
Obsolete
268
Inactive
Import
AML Search
Export Part Master Data
New Part
Default
Search part number, description or customer
Columns (8/14)
Part Number
Customer Name
Rev
Description
Part
00848-962-4138
00848 - KT Controls Ltd
B
CAP_CER-X7R 100NF 50V 0402
00455-1AB467205
00455 - Cerelogic Systems
C
PCB ASSY MAIN CONTROLLER REV C
00378-955-3951
00378 - Nokia Networks Oy
—
Standoff, 6mm M3 hex brass nickel
01204-1AB939586
01204 - Meridian Avionics
C
IC PWR MGMT BUCK 3A SYNC 4.5-18V
00378-750-6753
00378 - Nokia Networks Oy
B
Standoff, 6mm M3 hex brass nickel
01204-528-8362
01204 - Meridian Avionics
A
CAP_CER-X7R 100NF 50V 0402
00912-1AB289354
00912 - Halden Marine AS
A
CAP_CER-X7R 100NF 50V 0402
00848-1AB909252
00848 - KT Controls Ltd
—
SCREW M3×8 PAN HEAD SS A2
00378-1AB562000
00378 - Nokia Networks Oy
D
DIODE SCHOTTKY 40V 3A SMC
00912-540-4889
00912 - Halden Marine AS
A
CAP_CER-X7R 100NF 50V 0402
00912-1AB112828
00912 - Halden Marine AS
C
CAP_CER-X7R 100NF 50V 0402
About this prototype



Part C — Assets

C1 · Icons

One set. **20×20 viewBox, stroke only, no fill, one stroke weight.** A heavier line at a smaller size is how an icon set stops looking like one set.

An unknown name renders **nothing**, with the name left in the DOM, rather than falling back to a document glyph that hides the bug.

Default size	16px · `--vy-icon-sm`
In a grid row	15px inside a 26px button
Colour	`currentColor` — the icon takes the text colour around it
Decorative	`aria-hidden`, and the state it marks gets words of its own

The 45 names: `task · chat · log · quote · order · return · customer · requisition · supplier · factory · simulate · bom · transfer · adjust · pcb · tool · machine · invoice · journal · lock · people · template · job · plug · menu · globe · metadata · queue · doc · home · sell · buy · parts · stock · make · finance · insight · settings · chevron-down · chevron-right · eye · download · clip · trash · check`

Two directional glyphs, each with one job: **down** opens a menu below itself, **right** says "this opens somewhere else".

C2 · Illustrations

One drawing, three states. The product has exactly one illustrated character — a rocket — and it appears in three situations. That is deliberate: a reader who has met this craft on the way in should not have to learn a second picture to understand that it crashed.

Component: `src/ui/RocketLaunch.tsx` · one inline SVG, `viewBox="0 0 240 180"` · `data-state` selects the scene.

Rendered width	`clamp(240px, 18vw, 280px)`
Colour	Ten role tokens, B1.7 — never hex
Motion	CSS only. The illustration owns pixels and no state, so it can stop in a background tab and nothing is lost
Ids	Generated with `useId` — two instances on one page would otherwise share a clip path
Inline, not `<img>`	An `<img>` is a second request on the one screen whose job is to cover the first one, and it is its own document, where `var(--vy-brand)` resolves against nothing

C2.1 `launching` — the boot screen

The craft holds still in the middle of the frame and **the world moves**: the planet sinks a notch per boot stage, streaks run down past it. Nothing about a rocket drawn nose-up says "climbing" on its own.

Where	`BootGate` after sign-in; the `Suspense` fallback of a lazy route
Animation	Lift-off 0.7s · bob 1.6s · flame flicker 0.18s · streaks 0.8–1.3s
Reduced motion	No climb, no flicker, no streaks. The step bar still fills



Shown with its label and five-segment bar, which is what the boot screen actually is:



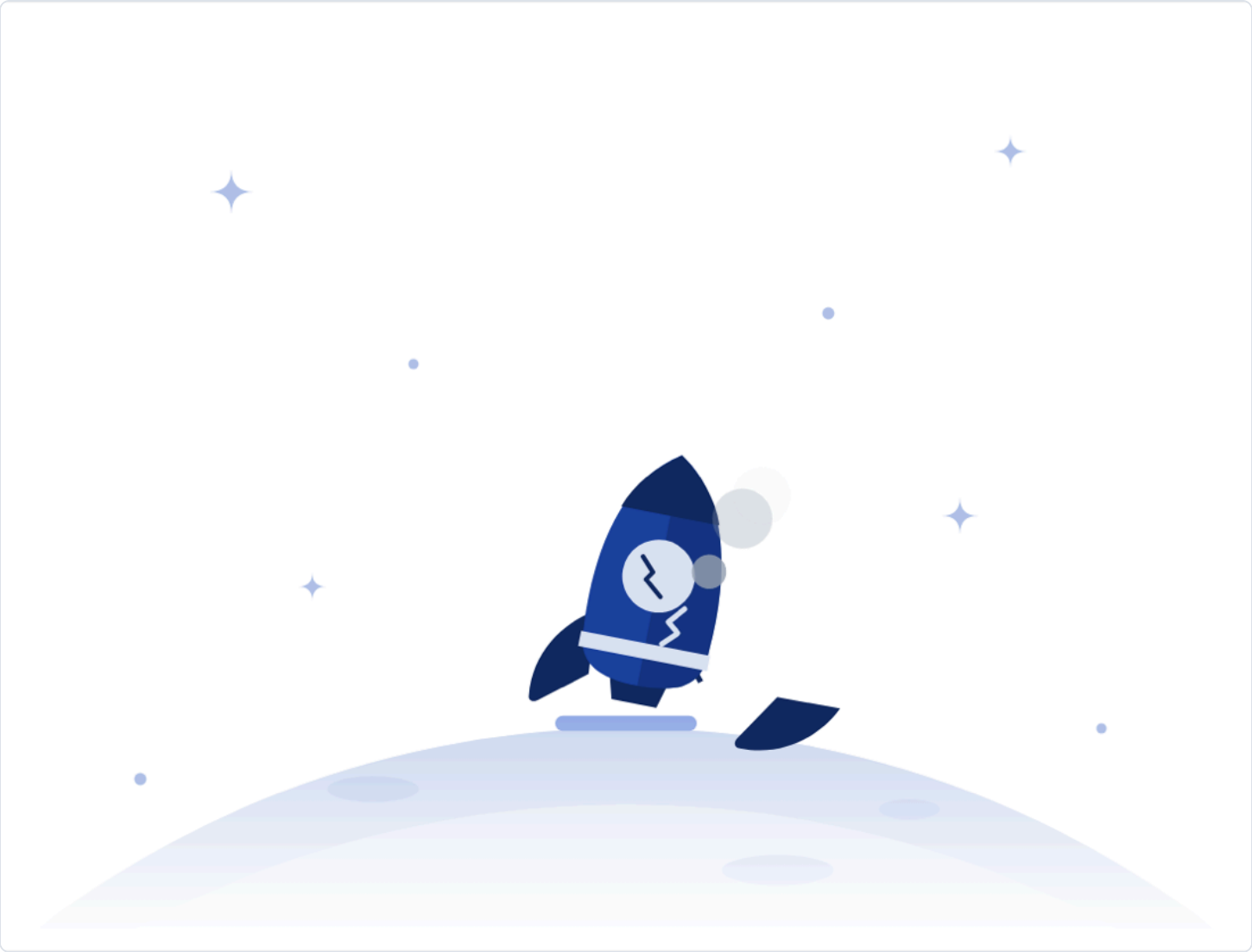
Restoring your settings

Specimen — stepping on a timer. In the product each stage waits on `CHANGE_FORMAT_CONFIG`, and the screen itself appears only if the boot takes longer than 0.4 s.

C2.2 `error` — the screen stopped loading

The same craft, back on the pad: tilted, cracked, one fin lying beside it, smoking. The failure is legible before the sentence under it is read.

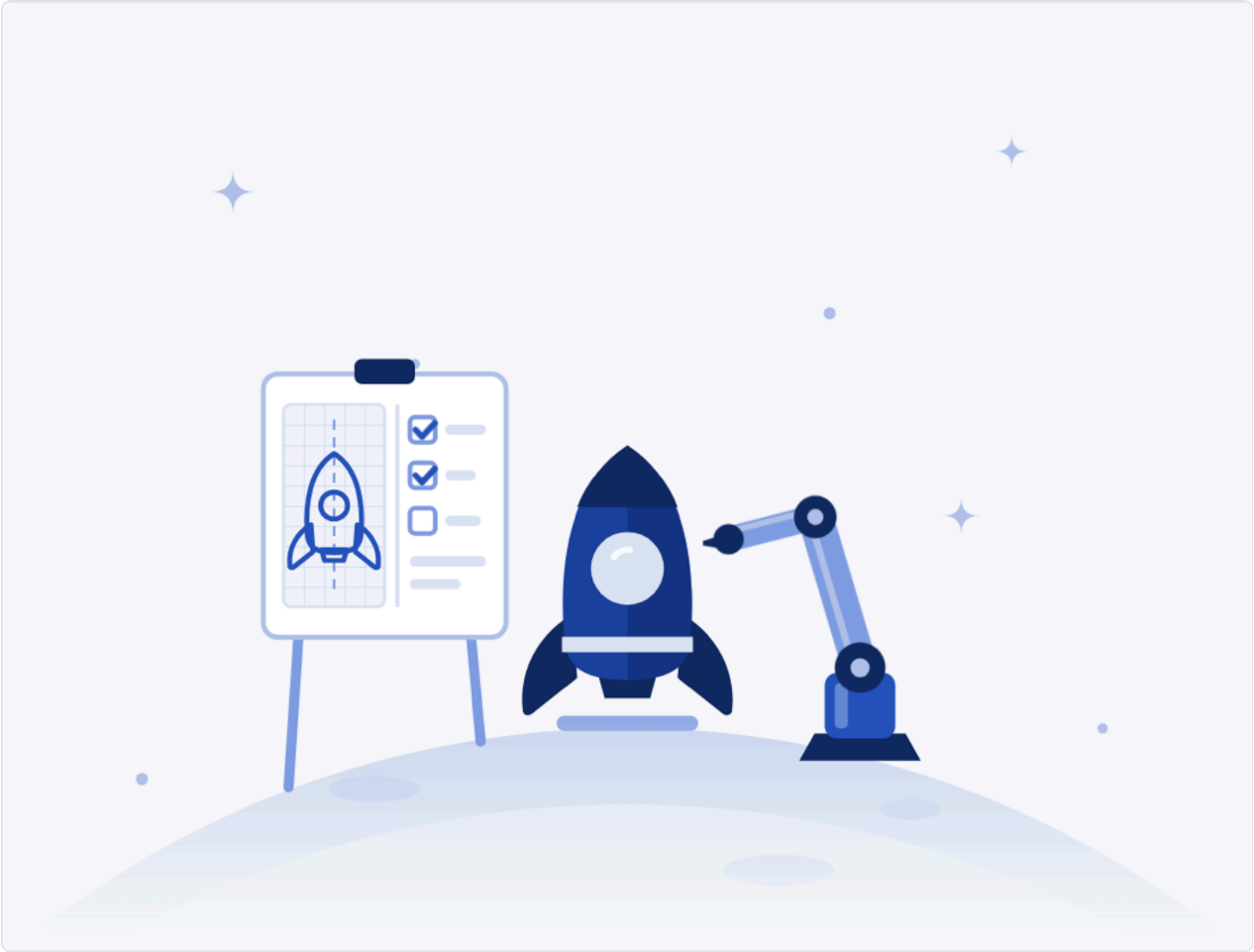
Where	`ErrorState`, which `ErrorBoundary` renders for any route that throws
Animation	Three smoke puffs, 2.4s, staggered
Reduced motion	Puffs held at 0.6 opacity — the smoke stays, the movement goes



C2.3 `building` — still on the launch pad

The craft **whole** on the pad, a robot arm welding it, a design doc on an easel whose checklist ticks off. Work in progress, not wreckage — and the difference has to be visible before the heading is read.

Where	`Placeholder`, which answers every route not yet built (~70 of 91)
Animation	Arm 4s · sparks only while the arm is holding · checklist 6s
Reduced motion	Sparks and the first two ticks held **on**, so the drawing still says "being built"

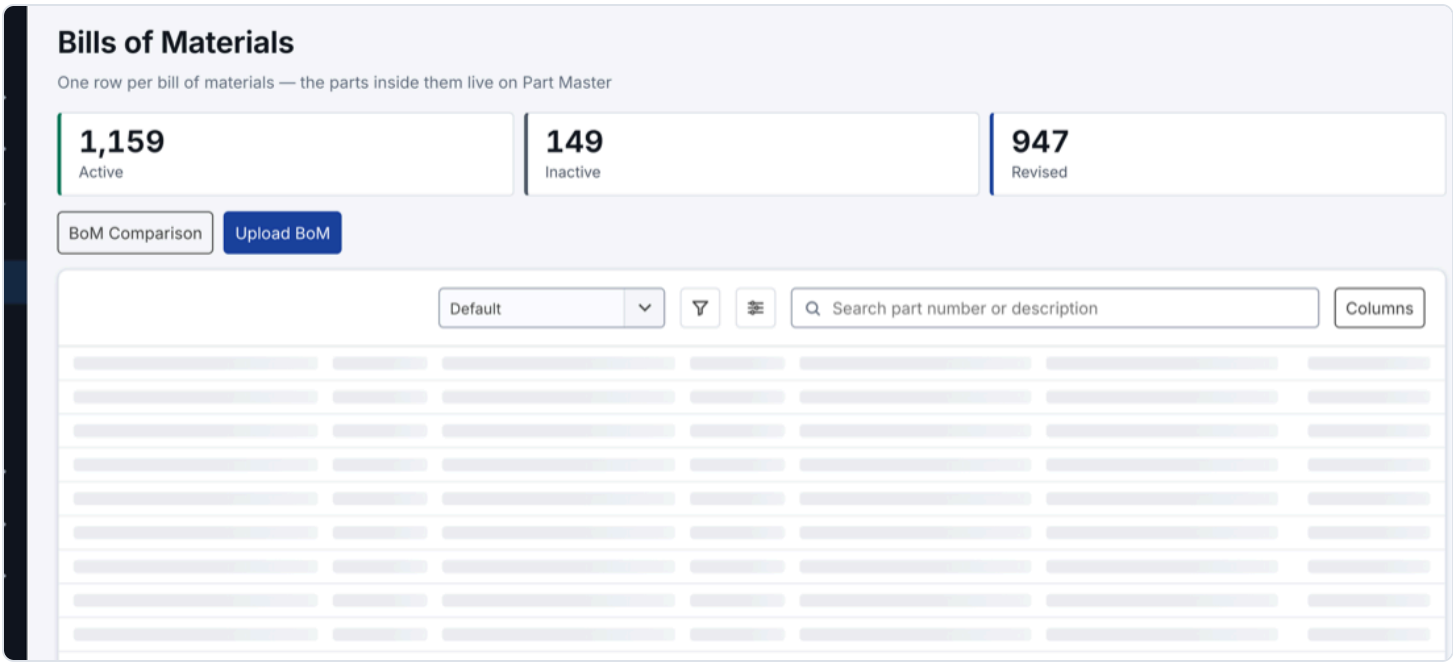


C2.4 Skeletons — the other kind of waiting

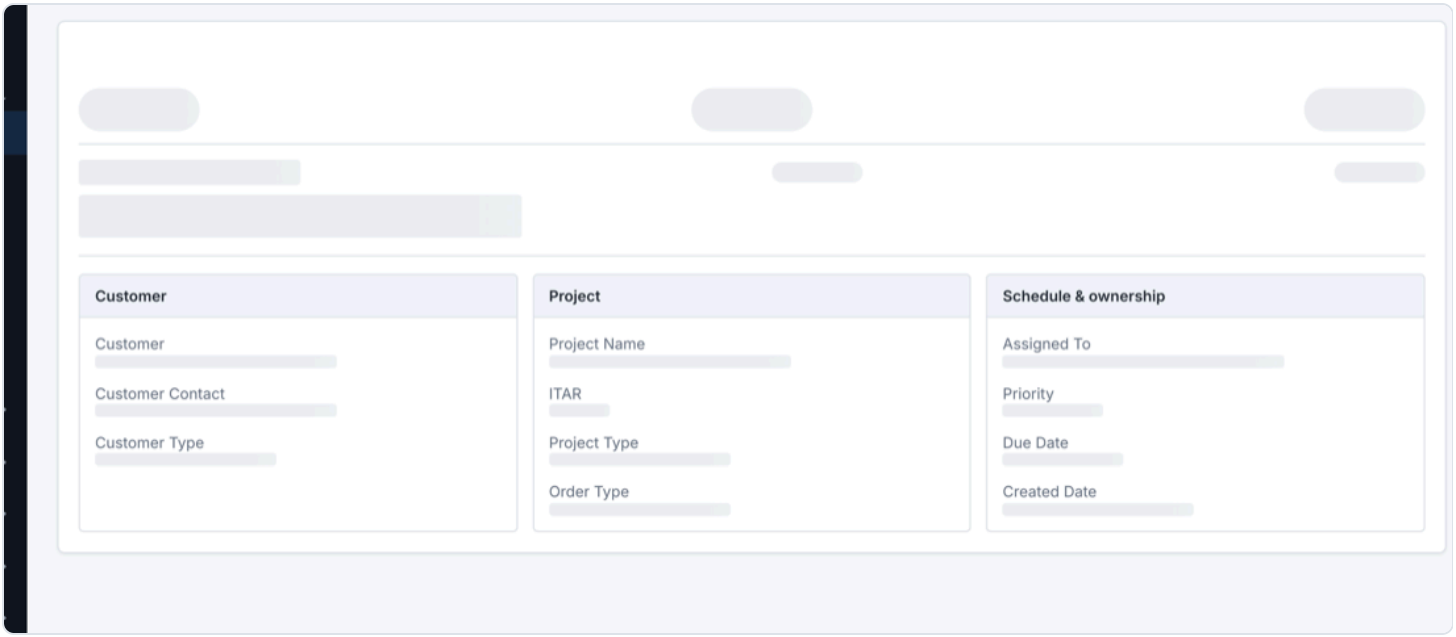
A skeleton answers *"what will this look like?"*. Once the user has already seen rows that question is answered, so a re-price or a second page **dims what is there** and puts an indicator over it instead (`BusyOverlay`).

One shimmer for both: `--vy-surface-sunken` → `--vy-surface-app` → `--vy-surface-sunken`, 200% background, 1.4s.

Grid skeleton — the real column roles, so the bars land where the data will:



Record skeleton — the real field labels, because they are known before the data is. Measured down to 1px of shift when the record arrives:



C2.5 Empty states

No illustration, on purpose. An empty state is a sentence and an action, and a picture there delays both.



C2.6 Exporting and regenerating

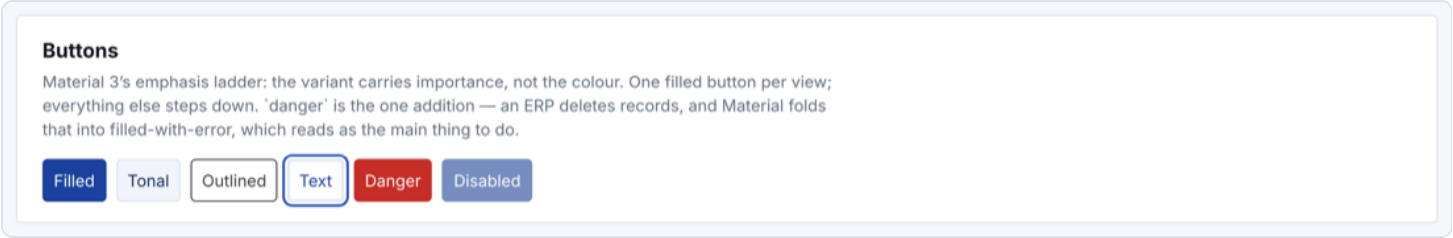
Live source	`src/ui/RocketLaunch.tsx`
Static SVGs	`docs/assets/rocket-journey/*.svg` — the **retired** horizontal set, kept because an earlier document still prints them. Not in the product
These images	Captured from the running app at 2×; the scripts are in `scripts/`

> **If an illustration needs to leave the codebase** — for a slide, a PDF — export the inline SVG and replace the `var(--vy-rocket-*)` references with the resolved hex from B1.7. Outside the app there is no token layer to resolve them.

Part D — Components and patterns

`src/ui` holds 33 files: the primitives, plus the icon set, the preference store and three hooks. Each wrapper has a reason, and the rule that produced them is the same every time — **the live system already ships KendoReact, so the question is never "what should this look like" but "what does Kendo already do, and what must we add".**

D1 · Buttons



The picture above is the live specimen with the keyboard on it: `Text` carries the focus ring every control in the system uses — 2px of `--vy-brand-boundary` offset by 2px, never a removed outline.

Variant	When
`filled`	**One per view.** The most likely next step
`outlined`	The default for everything else
`tonal`	A secondary action that still needs weight
`text`	Tertiary, inside dense rows
`danger`	Destructive — and it goes behind **More** ▾

Placement: all buttons on the **left** of a toolbar; a smart-button group or step indicator on the right; at most four visible, the rest behind `More ▾` — and the label is always the word *More* with a chevron, never `⋯`.

Cancel only in dialogs. In edit mode the word is **Discard Changes**.

`busy` **is not decoration.** The indicator says something is happening; the **disable** is what stops a second click spending a second marketplace request.

D2 · Fields

Two modes, three presentations

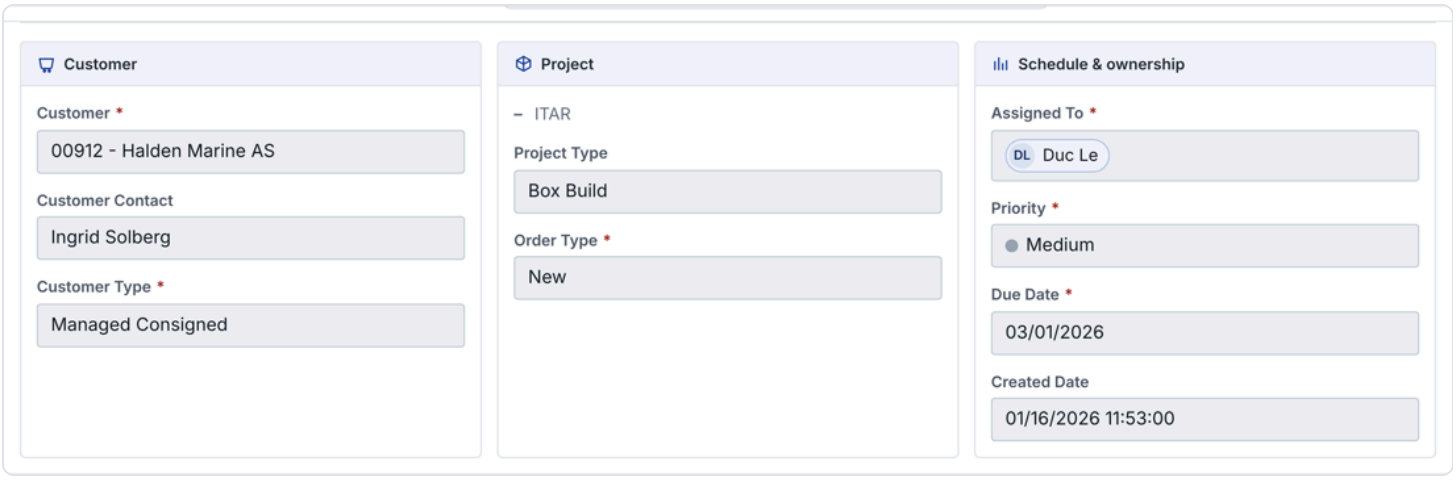
A record has **two modes** — view and edit. Inside edit mode a field is one of **two kinds**, and that is what the padlock is for. It is not a third state of a field; it is the mark that separates the two kinds, and it exists only where the distinction is live.

	Field the user can change	Field the user cannot
View mode	Sunken box, no padlock	Sunken box, no padlock
Edit mode	**White** input, `--vy-border-control`	**Sunken** box, **padlock**, and a short reason

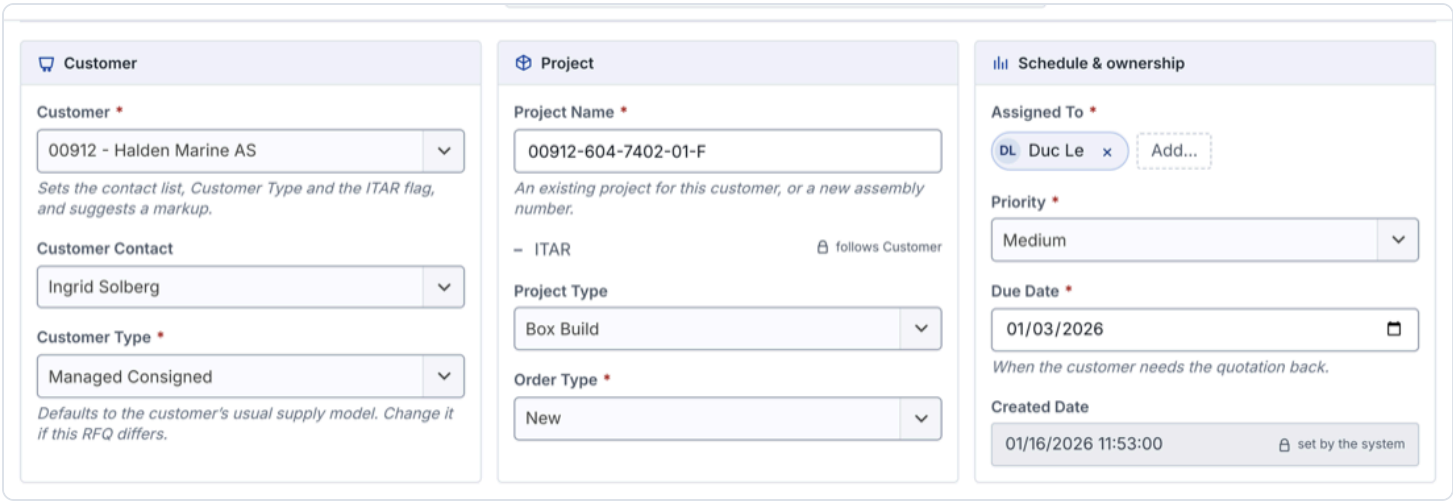
In view mode nothing is editable, so the distinction has nothing to say — a padlock on all twenty-seven fields marks none of them. The boxes are identical and the eye reads the values, not the chrome.

In edit mode the white tells you where you may type. Everything still in a grey box is a value this screen does not own, and the padlock is followed by the reason in two or three words, because a padlock says *that* a field is locked and cannot say *why*.

View mode — one box for every value:



Edit mode — white where you may type; grey, padlocked and explained where you may not. `Created Date` reads *set by the system*; `ITAR` reads *follows Customer*:



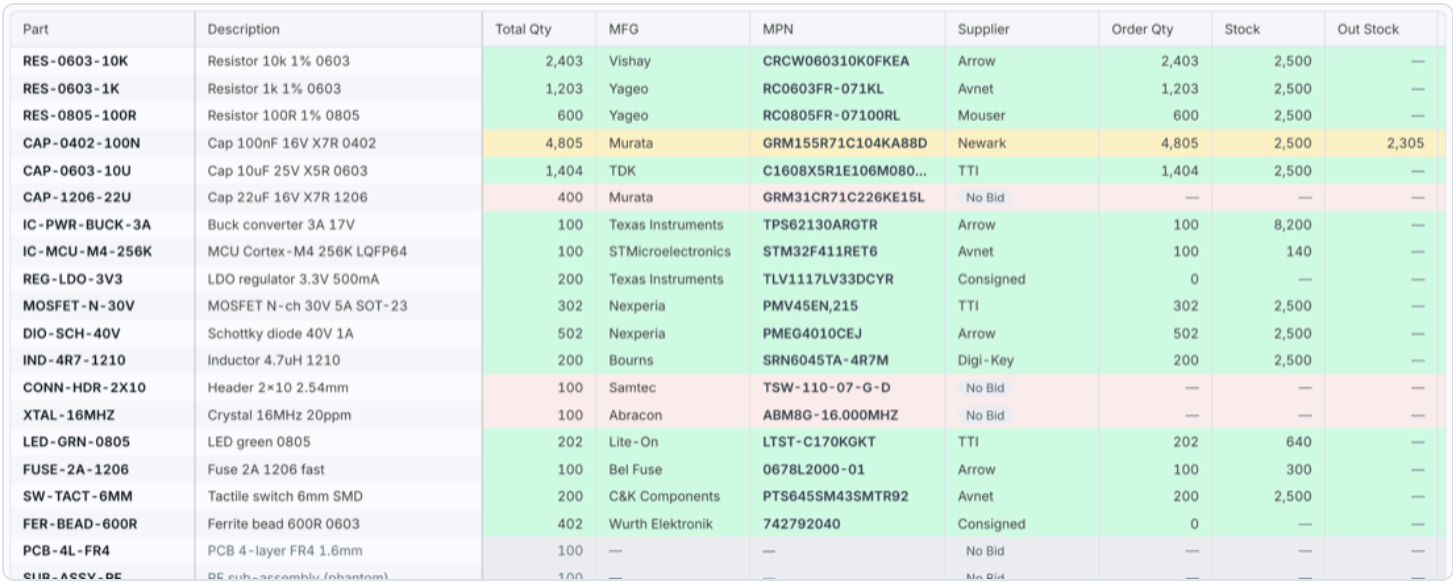
> **One exception, and it is the same rule.** A panel where *nothing* is editable keeps the grey box and drops the padlock — Run Quotation's setup strip on the last step, where the run is finished. A padlock on each of seven values says only what the panel already says.

The rest of a field

Part	Rule
Label	**Above** the value, never beside it. Title Case, 12px, 600
Required	A **red asterisk** right after the label, in view **and** edit mode
Optional	No mark. Never the word "(optional)"
Helper	Below the field, italic, at or below the label's size
Error	Red text below the field, **on blur** — not on every keystroke
Error *or* hint	Never both: two lines, each claiming to say what to do next
Hint	A **sibling** of the label, never inside it, or it joins the accessible name

D3 · Grids

A grid is where this system does most of its work, and the **row tone** is how it talks. A tone is a row's state in the quote, not a severity: green is covered by the chosen supplier, amber is short of the quantity, red is no bid, grey is excluded from the quote entirely.



Part	Description	Total Qty	MFG	MPN	Supplier	Order Qty	Stock	Out Stock
RES-0603-10K	Resistor 10k 1% 0603	2,403	Vishay	CRCW060310K0FKEA	Arrow	2,403	2,500	—
RES-0603-1K	Resistor 1k 1% 0603	1,203	Yageo	RC0603FR-071KL	Avnet	1,203	2,500	—
RES-0805-100R	Resistor 100R 1% 0805	600	Yageo	RC0805FR-07100RL	Mouser	600	2,500	—
CAP-0402-100N	Cap 100nF 16V X7R 0402	4,805	Murata	GRM155R71C104KA88D	Newark	4,805	2,500	2,305
CAP-0603-10U	Cap 10uF 25V X5R 0603	1,404	TDK	C1608X5R1E106M080...	TTI	1,404	2,500	—
CAP-1206-22U	Cap 22uF 16V X7R 1206	400	Murata	GRM31CR71C226KE15L	No Bid	—	—	—
IC-PWR-BUCK-3A	Buck converter 3A 17V	100	Texas Instruments	TPS62130ARGTR	Arrow	100	8,200	—
IC-MCU-M4-256K	MCU Cortex-M4 256K LQFP64	100	STMicroelectronics	STM32F411RET6	Avnet	100	140	—
REG-LDO-3V3	LDO regulator 3.3V 500mA	200	Texas Instruments	TLV1117LV33DCYR	Consigned	0	—	—
MOSFET-N-30V	MOSFET N-ch 30V 5A SOT-23	302	Nexperia	PMV45EN,215	TTI	302	2,500	—
DIO-SCH-40V	Schottky diode 40V 1A	502	Nexperia	PMEG4010CEJ	Arrow	502	2,500	—
IND-4R7-1210	Inductor 4.7uH 1210	200	Bourns	SRN6045TA-4R7M	Digi-Key	200	2,500	—
CONN-HDR-2X10	Header 2×10 2.54mm	100	Samtec	TSW-110-07-G-D	No Bid	—	—	—
XTAL-16MHZ	Crystal 16MHz 20ppm	100	Abracon	ABM8G-16.000MHZ	No Bid	—	—	—
LED-GRN-0805	LED green 0805	202	Lite-On	LTST-C170KGKT	TTI	202	640	—
FUSE-2A-1206	Fuse 2A 1206 fast	100	Bel Fuse	0678L2000-01	Arrow	100	300	—
SW-TACT-6MM	Tactile switch 6mm SMD	200	C&K Components	PTS645SM43SMTR92	Avnet	200	2,500	—
FER-BEAD-600R	Ferrite bead 600R 0603	402	Wurth Elektronik	742792040	Consigned	0	—	—
PCB-4L-FR4	PCB 4-layer FR4 1.6mm	100	—	—	No Bid	—	—	—
SUB-ASSY-RF	RF sub-assembly (phantom)	100	—	—	No Bid	—	—	—

Every tone is named under the grid it paints. A colour that only the author can decode is a colour that means nothing:

Rule	
Column chooser	On every grid with **3+ columns**
Frozen block	Declared as `locked`, so the chooser and the freeze cannot disagree
Zebra	By **record**, not by row — a record spanning two AML rows is one band
Totals	A grid the user sums by eye gets a totals row
Empty	A real empty state that says what to do, never "No records available"
Width	By role (B5.3); a deviation carries a `widthNote`

D4 · Dialogs

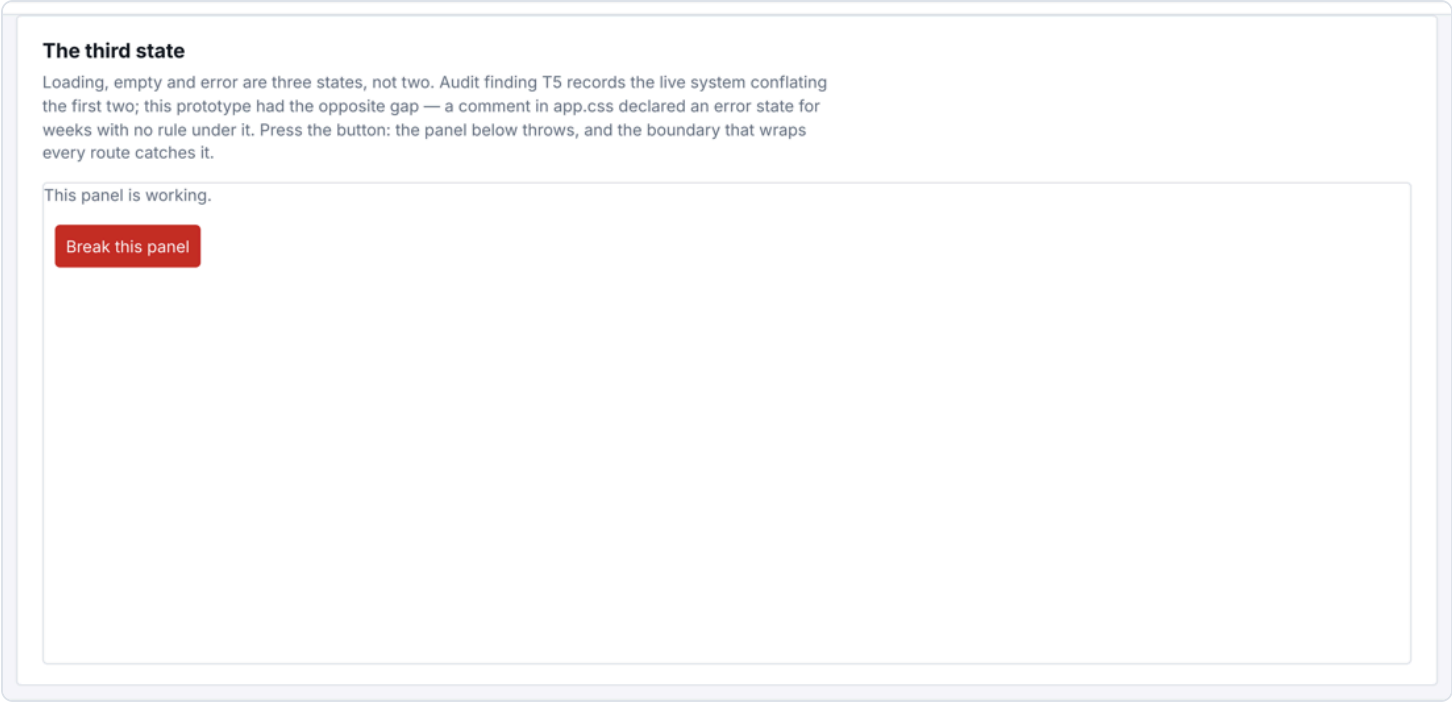
Sizes are in B5.4. Three further rules:

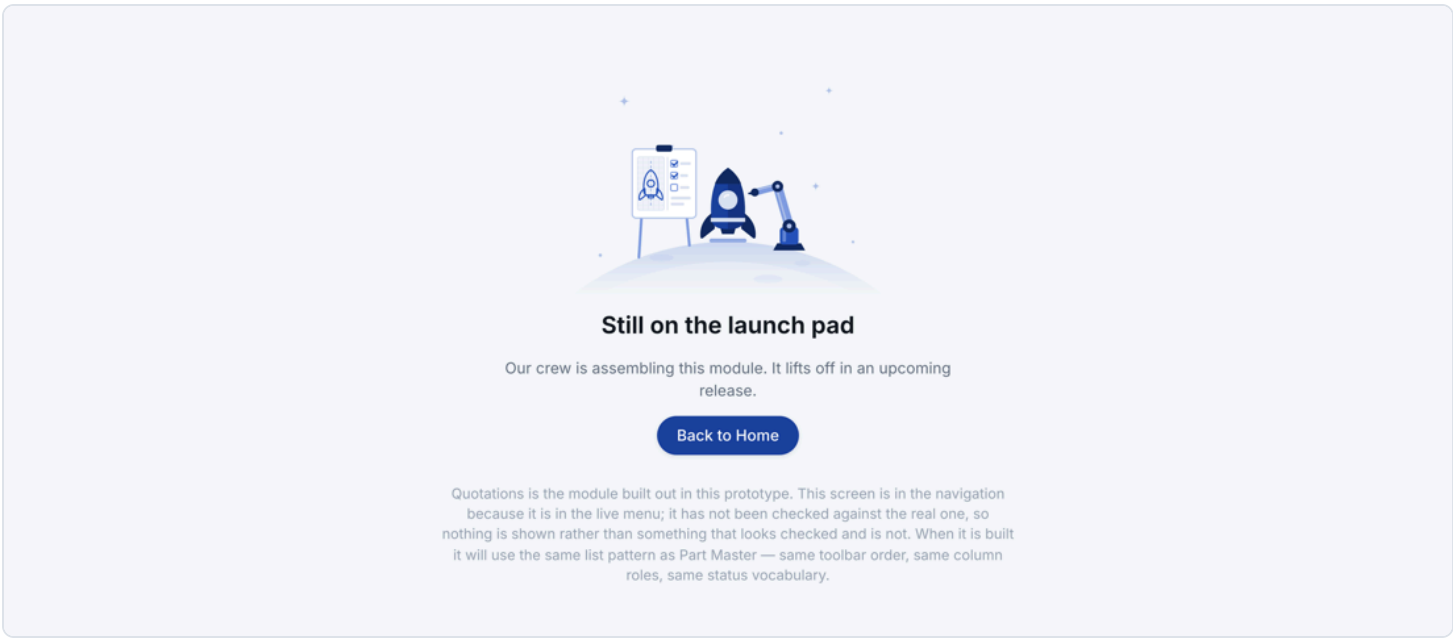
1. **Actions top-left** on a work surface, in a footer on a question. The way out goes on the left wherever the bar sits.
2. `lg` **and** `xl` **open maximised**; `md` never does. Those two are the work surfaces, and a 21-column grid in a 1180px window spends the screen it was given — while a confirmation filling a 27-inch screen is a worse answer than the one it replaced.
3. **Escape dismisses the innermost thing**, which the Select had to be taught by hand — Kendo's own handling closed the dialog from an open dropdown.

D5 · The four screen states

Loading, empty, error and not-yet-built are four different things and look it. Getting these wrong is the most common way a screen lies to its reader.

State	What the user sees	Actions	Illustration
Loading	Rocket climbing, planet sinking, five-segment bar	none	C2.1
Empty	A sentence saying what would fill it	the action that fills it	none
Error	Rocket on the pad: tilted, cracked, one fin off, smoking	Reload · Back to Home · Send a support ticket · Error ID + Copy	C2.2
Under development	Rocket whole on the pad, arm welding, plan ticking off	Back to Home. **No Reload — nothing is broken**	C2.3





The error state's three slots are the three questions a person actually has: *what failed · what it means for me · what I can do next.* Only the first is generic.

The error id is minted where the error is caught, not where it is drawn — minted in render, pressing Copy would hand the user a different number for the same failure.

D6 · Words

Casing	B2 — Title Case for labels, sentence case for prose
A control says what happens	`Publish`, then a toast saying `Published`
An error says what to do	Not "Something went wrong"
Nothing is "not implemented" silently	A toast says what it *would* do
A derived figure says it is derived	*"Derived from this run — the live system takes them from the customer's purchase history"*
A session-only change says so	*"Added in this session… nothing here is saved"*

Part E — Quality

E1 · Accessibility

Measured, not asserted. The full sweep is in `contrast-sweep.md`; the keyboard work is in `focus-order.md`.

Requirement	Threshold	Status
Text contrast — WCAG 1.4.3	4.5:1 normal · 3:1 large (≥24px, or ≥18.66px at ≥700)	**78 combinations measured, 0 failures**
Non-text contrast — WCAG 1.4.11	3:1 for control boundaries	Three controls failed at 1.30 / 1.64 / 2.55 · fixed by `--vy-border-control` at **3.06:1**
Target size — WCAG 2.5.5	44×44	Enforced under `pointer: coarse`
Reduced motion	respected	B6

Why the contrast harness can be trusted: colours are read through a **canvas pixel** rather than parsed from strings, because Kendo emits `oklch()`, which no regex over `getComputedStyle` survives. Backgrounds are composited up the ancestor chain. Before the results were believed, a `#bbbbbb`-on-white element was planted in the page and caught at **1.92:1** — a sweep that reports no failures is only worth reading if it can prove it would have found one.

Four rules that are easy to get wrong and are already settled:

- A **status is a word and a colour**, never a colour alone.
- A **hint is a sibling of the label**, never inside it.
- `role="alert"` **for an error,** `role="status"` **for a loading state.** One interrupts, the other does not.
- A **decorative glyph is** `aria-hidden` and the state it marks gets words of its own — a padlock plus a screen-reader-only "read-only".

E2 · KendoReact

The live system already ships KendoReact 16. The bridge is three files:

File	Job
`kendo-subset.scss`	Compiles only the Kendo components actually used
`_kendo-size.scss`	**Configures** Kendo's size scale — does not override it
`kendo-bridge.css`	Maps Kendo's variables onto ours, and fixes what cannot be configured

Specificity is the trap. `.k-button.k-button-solid.k-button-base.k-focus` weighs (0,4,0), so a one-class rule of ours loses silently. The bridge prefixes with `html[data-kendo-bridged]` rather than repeating a class five times.

Two Kendo behaviours worth knowing before debugging a component:

- **Kendo disables with a `.k-disabled` CLASS, not the `:disabled` attribute.** A plain `<input disabled class="k-radio">` renders identically to an enabled one.
- **KendoReact 16 locks grid columns with sticky cells in one table** (`position: sticky`), not with two panes.

Part F — Handover

F1 · File map

Path	Contents
`src/theme/tokens.css`	**Every token.** Light, plus the two dark blocks
`src/theme/tokens.ts`	The values TypeScript needs: column widths, dialog widths
`src/theme/md3.css`	Motion, focus ring, the Material 3 structure we kept
`src/theme/base.css`	Reset and element defaults
`src/theme/components.css`	Component rules
`src/theme/app.css`	Screen and layout rules
`src/theme/responsive.css`	Every breakpoint, in one file
`src/theme/kendo-*.{css,scss}`	The Kendo bridge
`src/ui/`	The primitives
`src/ui/RocketLaunch.tsx`	The illustration, all three states
`/design-system`	The living specimen page
`docs/assets/design-system/`	The images in this document

F2 · How to change something

A colour token

1. Add or change it in **all three** blocks of `tokens.css` (B1.8).
2. If it lands on text or a control boundary, re-run the contrast sweep.
3. `npm run check:fast` — lint, stylesheet check, typecheck, build.

A component

1. Check whether Kendo already does it (E2).
2. Build it in `src/ui`, referencing semantic tokens only.
3. Add a specimen to `/design-system` if it has states worth seeing.

4. Cover every state: default, hover, focus, active, disabled, busy, error.

A new screen

1. Which of the four states (D5) can it be in? Build all of them.
2. Column roles, not pixel widths.
3. Check it at 1180, 980 and 820, in both themes.

What CI runs on every push: `render:check` · `touch:check` · `mobile:check` · `focus:check` · `hover:check`.

F3 · Governance

Source of truth	`src/theme/` and the components in `src/ui`. This file describes them; it does not define them
Where a decision is recorded	In the code, at the rule it governs — and in a dated note under `docs/` when a reader would otherwise have to re-derive it
Judgement	Allowed where nothing else answers, and **labelled as judgement** where it is used
Versioning	`v1.0` is this handover. A token or rule change is a minor version; a change that requires screens to be rebuilt is a major one

Three rules that keep this document true:

1. **A magic number needs a note** saying what was measured.
2. **A comment states what was measured, not what was intended.** Three corrections in this project came from a comment describing behaviour the code did not have.
3. **The living page beats the document.** `/design-system` renders the real components; if the two disagree, the page is right and this file is stale.

F4 · What a handover should contain

The shape of this document was checked against current practice rather than invented. The recurring list across published checklists is: **design tokens** with their values · **component specs** with every state and variant · **usage rules** with do and don't · **accessibility** with the WCAG thresholds named · **code**

examples · **responsive** behaviour with the breakpoints stated · **motion** with duration, easing and a reduced-motion alternative · and **governance** — ownership, a contribution process, versioning.

Several sources also name the one failure mode this file is most exposed to: documentation drifting away from the live code once the team moves on. That is why F3 makes the specimen page the tie-breaker rather than this document.

Consulted 3 November 2026: Magic Patterns · Figr checklist · Supernova · Docsie · Eleken · UX Planet. These are vendor blogs and community checklists, not a standard; they agree on the core set and differ on depth.

F5 · Source index

Document	What it settles
`contrast-sweep.md`	Every measured contrast ratio, and the method
`focus-order.md`	Keyboard order and the focus ring
`modal-patterns.md`	Dialog sizes, action placement, dismissal
`table-patterns.md`	Grid behaviour, column roles, density
`motion-loading.md`	Every loading and transition decision
`kendo-control-scale-18oct.md`	The 33-rule Kendo size audit
`working-agreement.md`	Engineering rules, including the magic-number exceptions

Captured from the running build at 2× on 3 November 2026. Images live in `docs/assets/design-system/` *and can be regenerated from* `/design-system` *and the product screens they came from.*